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Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following pairs of regular expressions are equivalent?

Answer (Please select your correct option)

- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x^+ and x^*
- x^+ and x^*x^+

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Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

Answer (Please select your correct option)

- All
- No
- One
- Two

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Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Pumping lemma is generally used to prove that:

Answer (Please select your correct option)

- A given language is infinite
- A given language is not regular
- Whether two given regular expressions of a regular language are equivalent or not
- A given language is finite

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Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the language L of strings, defined over Σ = {a,b}, ending in a

Answer (Please select your correct option)

- There are finite many classes generated by L, so L is regular
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If L1 and L2 are two regular languages, then _____ is/are also regular.

Answer (Please select your correct option)

- Both $L1 \cup L2$, $L1 \cap L2$
- $L1^c \cap L2$
- $L1 \cup L2$
- $L1 \cap L2$

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Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Even-palindrome is a _____ language

Answer (Please select your correct option)

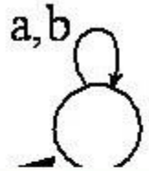
- non-regular
- regular
- regular but infinite
- regular but finite

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Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA accepts _____ language.



Answer (Please select your correct option)

- Finite
- Infinite
- No
- All

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Question No : 5 of 52 Marks: 1 (Budgeted Time 1 Min)

Using tree structure, initial state can be represented by____

Answer (Please select your correct option)

- An arrow
- +
- -
- *

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Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If an FA accepts no string, then language is denoted by _____

Answer (Please select your correct option)

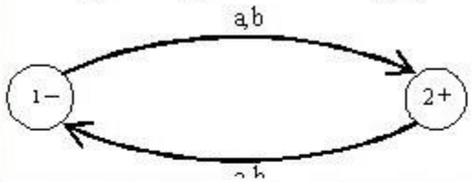
- {}
- Ø
- Both { } and Ø
- Null

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Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA generates the language having strings of _____



Answer (Please select your correct option)

- ODD length
- EVEN length
- Equal number of a's and b's
- Odd number of a's and b's

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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If L1 and L2 are expressed by r1 and r2, respectively then the language expressed by (r1)(r2) will be _____

Answer (Please select your correct option)

- Regular
- Ir-regular
- Undermined
- Such language does not exist

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement is true?

Answer (Please select your correct option)

- All words are strings
- All strings are words
- Both are always same
- All letter are string

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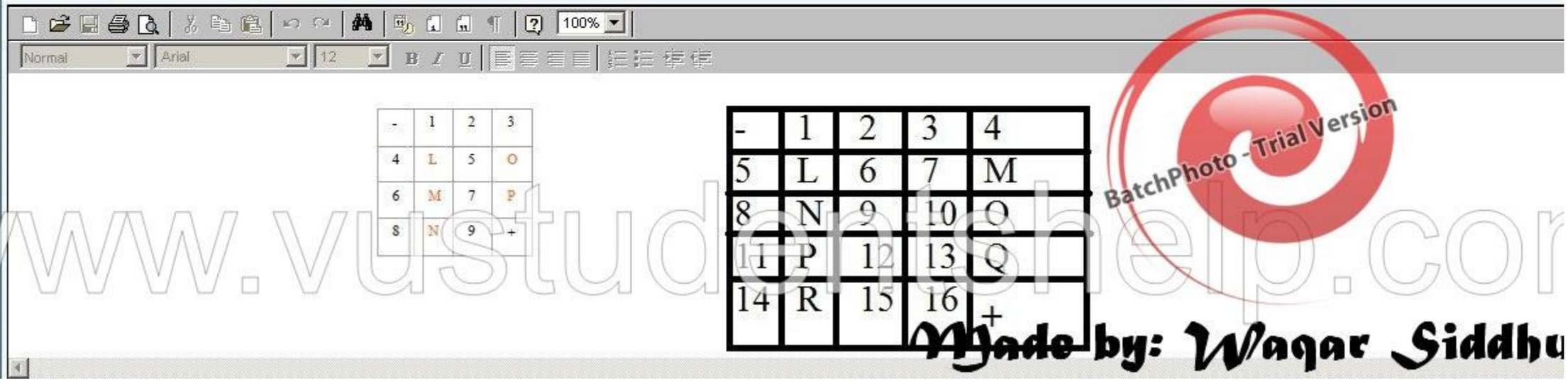
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Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Suppose a Maze of 20 boxes; 1 box for initial and 1 for final state. 6 boxes are for non-moveable states while the rest can be used for transitions. Draw a Nondeterministic Finite Automaton (NFA) fo the given Maze.

Marks [2.5+2.5]

Answer (Please click here to Add Answer)



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Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Derive Chomsky Normal Form (CNF) corresponding to the Context Free Grammar (CFG) given below:

$A \rightarrow BAB \mid B \mid \Lambda$

$B \rightarrow 00 \mid \Lambda$

Answer (Please click here to Add Answer)

A-> BAB|B|^

B->00|^

Null prod: A->BAB|B

Nullable prod: B->^

Remove null prod. Just

A->BAB|AB|BA|BB|A|B

A->B|A is the unit prod as B->00 so

A->BAB|AB|BA|BB|A|B|00

B->00

Interduce new C->BA

C->BA

So CNF will be:

A->CB|AB|BB|A|B|C00

B->00

C->BA

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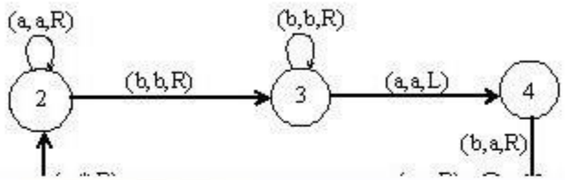
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Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the following TM



Answer (Please click here to Add Answer)

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above TM. It can also be observed that the above TM accepts the non-CFL {an bn an}.

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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the language L which is EVEN-EVEN, defined over Σ = {a,b}. In how many classes does L may partition Σ*. Explain briefly.

Answer (Please click here to Add Answer)

four classes

Consider the language L which is EVEN-EVEN, defined over Σ = {a,b}. It can be observed that L partitions Σ* into the following four classes

C1 = set of all strings with even number of a's and odd number of b's.

C2 = set of all strings with odd number of a's and odd number of b's.

C3 = set of all strings with odd number of a's and even number of b's.

C4 = set of all strings with even number of a's and even number of b's.

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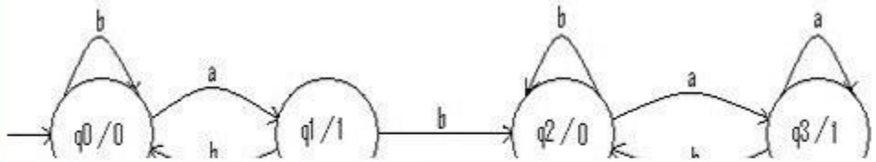
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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Consider the following Moore machine:



Answer (Please click here to Add Answer)

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Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Illustrate Decidable Problem with the help of an example.

Marks [1+2]

Answer (Please click here to Add Answer)

Decidable problem

A problem that has decision procedure is called decidable problem *e.g.* the following problems

The two regular expressions define the same language.

The two FAs are equivalent.

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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write three different parsing techniques.

Answer (Please click here to Add Answer)

There are different parsing techniques, containing the following three
Top down parsing.
Bottom up parsing.
Parsing technique for particular grammar of arithmetic expression.

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Question No : 45 of 52 **Marks: 3 (Budgeted Time 6 Min)**

What is meant by the following statement ?

"LANGUAGE IS CLOSED"

Answer (Please click here to Add Answer)

Closed w.r.t. concatenation i.e.the language expressed by RE of type r*.

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Question No : 44 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Consider live productions to be applied during derivation of a word. In your point of view what will be the ultimate result?

Answer (Please click here to Add Answer)

Live production: A production of the form nonterminal → string of two nonterminals is called a live production. result will be self embended

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Question No : 43 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Consider the language defined by the following regular expression:

b(a + b)*a

What will be the corresponding Context Free Grammar (CFG)?

Answer (Please click here to Add Answer)

S->bBa

B->Ba|Bb|a|b

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Question No : 42 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Explain the language L of string, defined over $\sum = \{0,1\}$, having double 0's or double 1's?

Answer (Please click here to Add Answer)

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$(0+1)^* (00 + 11) (0+1)^*$

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Question No : 41 of 52 **Marks: 2 (Budgeted Time 4 Min)**

What is meant by "Left most Derivation" in context of CFG?

Answer (Please click here to Add Answer)

production is applied to the left most

nonterminal in the working string, is said to be ***left most derivation***

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Question No : 40 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Let us consider the following Turing machine (™):



Answer (Please select your correct option)

- fffgggg
- fffggggf
- fffggg
- ffffggg

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Question No : 39 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S---->01A

A---->10B

B---->0A|11

Answer (Please select your correct option)

- 011011 not sure my thinking.... (y)
- 011011011
- 011001011
- 011001001011

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Question No : 38 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following states do not have outgoing edges:

Answer (Please select your correct option)

- Start
- Read
- Pop
- Accept

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

A read state can have _____ outgoing edge/ edges.

Answer (Please select your correct option)

- 1
- 2
- 3
- Any number of

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S → SX|XY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

- 00100
- 0010000
- 001
- 00010

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular

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Question No : 34 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following theorem partitions Σ^* into distinct classes?

Answer (Please select your correct option)

- Decidability theorem
- Myhill Nerode theorem
- Kleen's theorem
- Emptiness theorem

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The language of $a^n b^n$ can be expressed by:

Answer (Please select your correct option)

- FA
- TG
- GTG
- PDA

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

We can have more than one parse trees for a given string in case of _____ Context Free Grammar (CFG).

Answer (Please select your correct option)

- vague
- ambiguous
- hazy
- simple

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Question No : 31 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

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Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In _____ there are Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate, when the HALT states are entered.

Answer (Please select your correct option)

- TG
- Turing Machine
- GTG
- Finite Automata

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Can a turing machine's head ever be in the same location in two successive steps?

Answer (Please select your correct option)

- Yes
- Yes but only in finite languages
- No not sure
- Yes but only in infinite langauges

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If a given Context Free grammar (CFG) contains a production of the form S→Λ, then _____

Answer (Please select your correct option)

- The corresponding CFL is not empty.
- The corresponding CFL is empty.
- The corresponding CFL is not empty if the language is regular
- The corresponding CFL is empty and non regular.

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Question No : 27 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Left hand side of a production in Conext free grammer (CFG) consists of _____.

Answer (Please select your correct option)

- One terminal
- More than one terminal
- One non-terminal
- Terminals and non-terminals

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ machine helps in building a machine that can perform the addition of binary numbers.

Answer (Please select your correct option)

- Incrementing
- Complementing
- Decrementing
- Multiplication

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is RE for a language defined over Σ={a,b} having words starting with a:

Answer (Please select your correct option)

- (a + b)*
- aa(a + b)+
- a(a + b)*
- a*(a + b)

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)

If r1 and r2 are regular expressions then (r1) (r2) is a _____ .

Answer (Please select your correct option)

- RE
- TG
- FA
- GTG

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL)
- Regular
- Finite

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Question No : 20 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Σ= {a,b} then string abbabb has length equal to _____.

Answer (Please select your correct option)

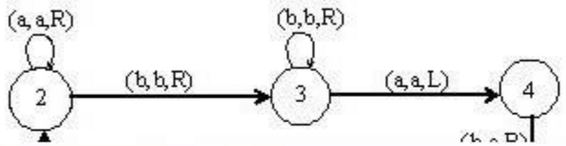
- Three
- Four
- Five
- Six

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Question No : 19 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following TM



Answer (Please select your correct option)

- Above TM accepts the non-CFL {a b c}.
- Above TM accepts the non-CFL $\{a^n b^n a^n\}$.
- Above TM accepts the non-CFL $\{a^n b^{n+2} a^n\}$.
- Above TM accepts the non-CFL $\{a^{n+2} b^n a^n\}$.

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

We can also represent an FA using different states e.g. Accept state, Reject state, Read State etc.
The _____ state behaves as final state of an FA.

Answer (Please select your correct option)

- Accept
- Pop
- Push
- Reject

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Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following statements is INCORRECT:

Answer (Please select your correct option)

- (a+b)*aa(a+b)* generates Regular language.
- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

There exist _____ derivations of a certain word in ambiguous grammar.

Answer (Please select your correct option)

- more than one
- Exactly one
- exactly two
- no

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Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement is correct?

Answer (Please select your correct option)

- The length of output string is greater than length of input string in moore machine.
- The length of output string is greater than length of input string in mealy machine.
- The length of output string is equal to length of input string in moore machine.
- The length of output string is less than length of input string in mealy machine.

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Question No : 13 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are not equivalent?

Answer (Please select your correct option)

- (ab)* and a*b*
- x(xx)* and (xx)*x
- x^+ and x^*x^+
- x(x)* and (x)*x

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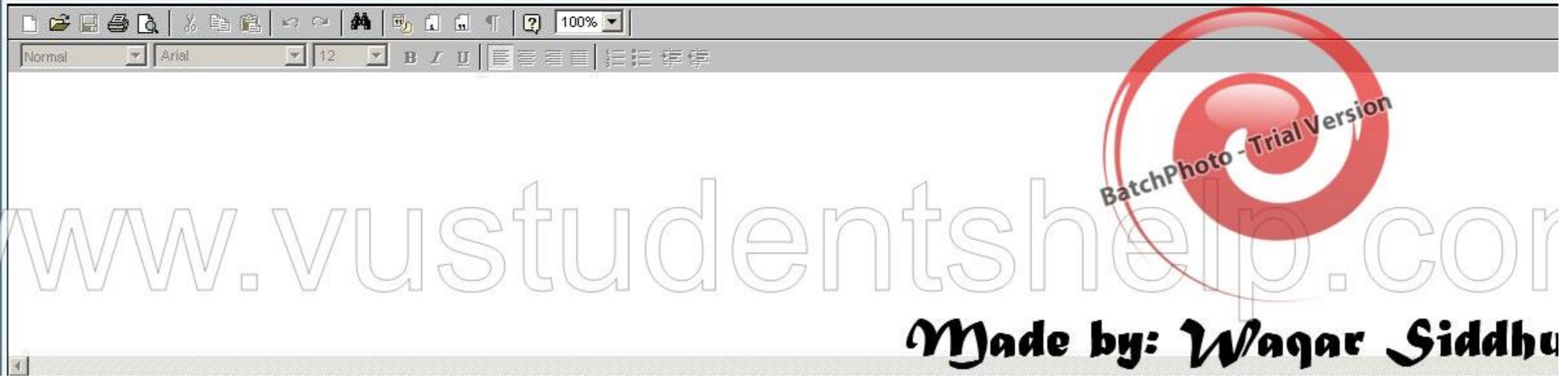
Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

a. Construct Context Free Grammar (CFG) for the language defined by the regular expression:

(ab + ba)*(a + b)*

b. Remove null productions from the CFG constructed in part i.

Answer (Please click here to Add Answer)



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Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Using suitable examples, differentiate between Kleene star closure and plus operation?

Answer (Please click here to Add Answer)

Kleene Star:-Given Σ, then the Kleene Star Closure of the alphabet Σ, denoted by Σ*, is the collection of all strings defined over Σ, including Λ

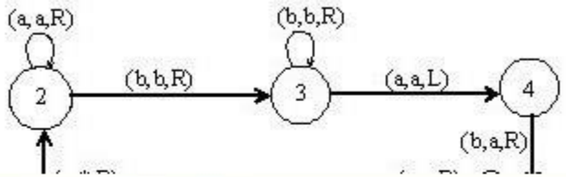
PLUS Operation (+)
Plus Operation is same as Kleene Star Closure except that it does not generate Λ (null string), automatically.

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Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the following TM



Answer (Please click here to Add Answer)

The string aaabbbaaa can be observed to be accepted by the above TM. It can also be observed that the above

TM accepts the non-CFL {an bn an}

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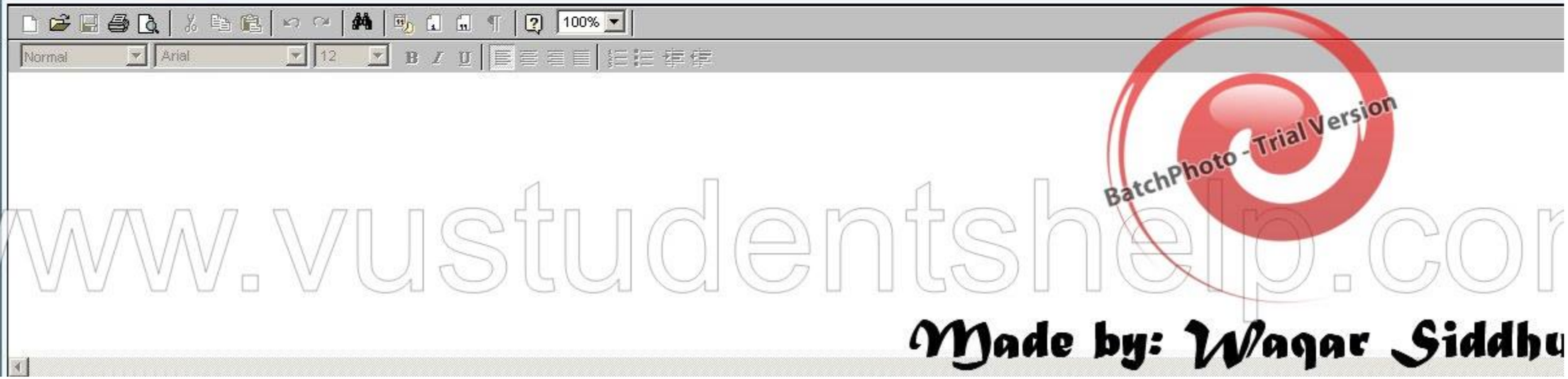
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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Write the CFG for EVEN-EVEN language?

S-> ^|SS|XS|YSY|
X->aa|bb
Y->ab|ba

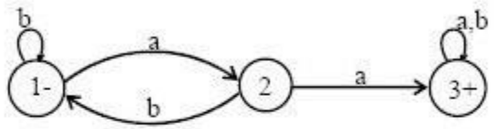
Answer (Please click here to Add Answer)



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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Let us consider the following Finite Automaton (FA):



Answer (Please click here to Add Answer)

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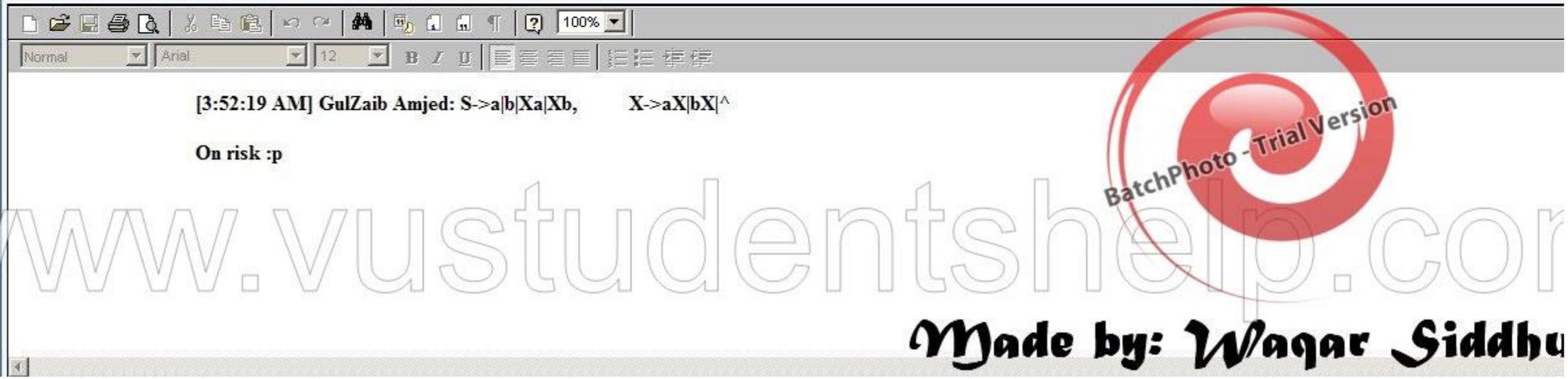
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Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Write down Context Free Grammar (CFG) for the language of strings expressed by the following regular expression:

$(a + b)^*(a + b)^+$

Answer (Please click here to Add Answer)



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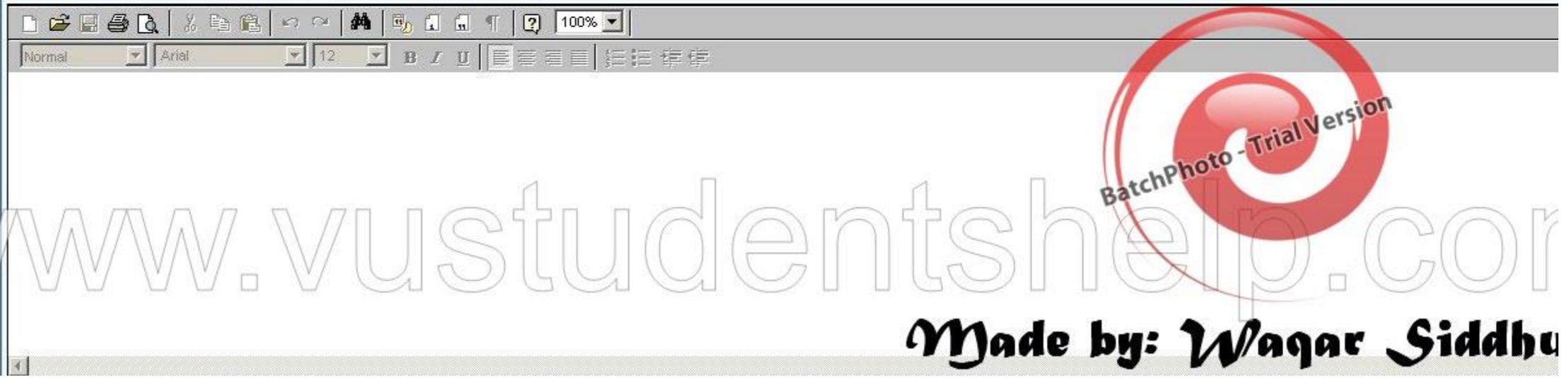
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Question No : 46 of 52 **Marks: 3 (Budgeted Time 6 Min)**

What does the following arbitrary summary table show?

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_0$	$READ_3$	b	b	abb	11

Answer (Please click here to Add Answer)



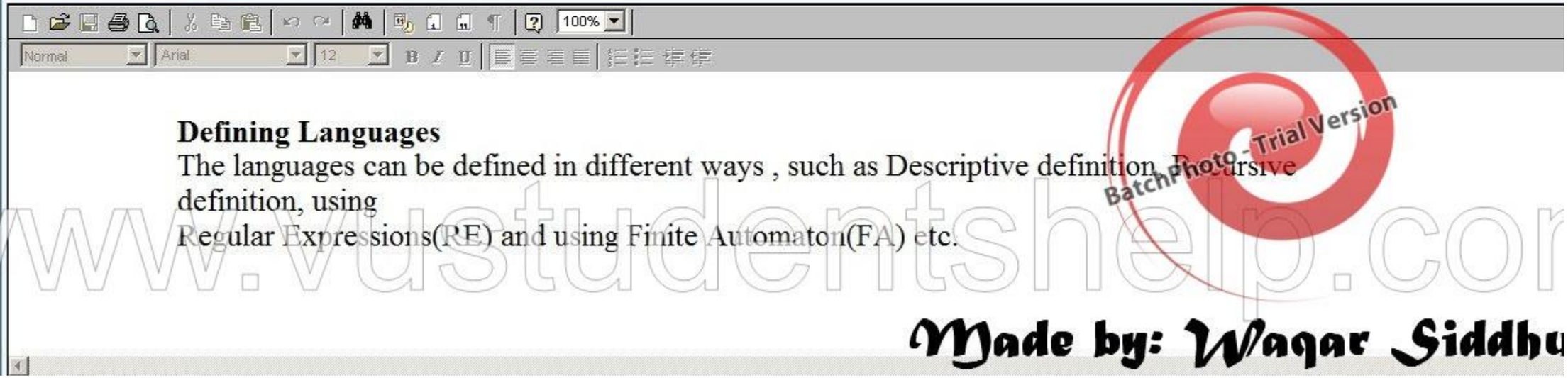
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What are the steps of Recursive Definition of Languages?

Answer (Please click here to Add Answer)



Defining Languages

The languages can be defined in different ways , such as Descriptive definition, Recursive definition, using

Regular Expressions(RE) and using Finite Automaton(FA) etc.

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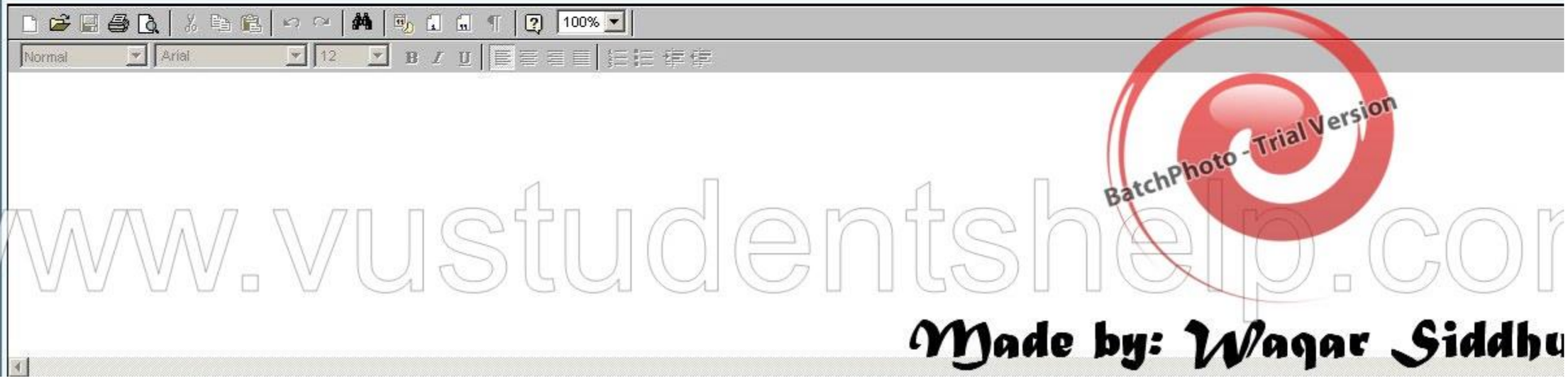
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Question No : 44 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Write down regular expression for a language L of strings defined over Σ = {a, b} that neither ends in "a" nor with "b".

Answer (Please click here to Add Answer)



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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by Row language?

Answer (Please click here to Add Answer)

, The alphabets consists of letters which are the rows in the summary table.

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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the uses of Push Down Automata (PDA) in computing?

Answer (Please click here to Add Answer)

The PDA is used in theories about what can be computed by machines. PDA described with a formal grammar, it can be used in parser design. The deterministic pushdown automaton can handle all deterministic context-free languages while the nondeterministic version can handle all context-free languages.

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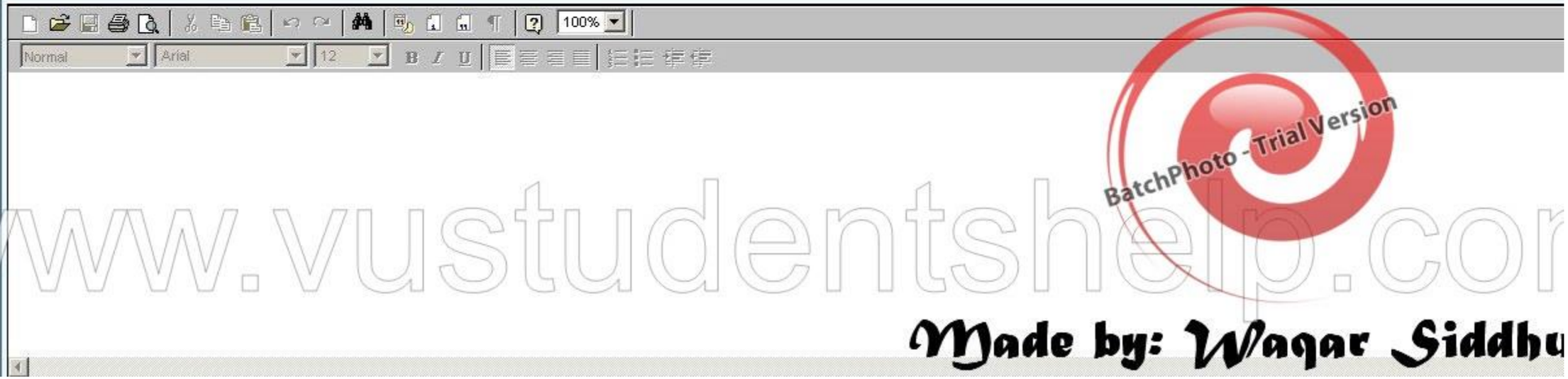
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Question No : 41 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Write alternative form of this production.

$S \rightarrow aS,\ S \rightarrow bS,\ S \rightarrow \Lambda$

Answer (Please click here to Add Answer)



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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is always Deterministic.

Answer (Please select you correct option)

- Finite Automaton
- Transition Graph
- Generalize Transition Graph
- Non-deterministic finite automaton

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Question No : 39 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A nonterminal is said to be self-embedded, if in a given derivation of a word, it ever occurs as a tree _____ of itself.

Answer (Please select your correct option)

- ascendant
- descendant
- top
- bottom

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Concatenation of r1 and r2 will be shown as:

Answer (Please select your correct option)

- r1+r2
- r2+r1
- r2r1
- r1r2

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a regular Context Free Grammar:

Answer (Please select your correct option)

- S→abS | baS | ^
- S→aSb | baS |^
- S→abS | bSa | ^
- S→aSb | Sa | ^

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S → SX|XY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

- 001
- 00010
- 0010000
- 00100

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S ---> SX|XY
X ---> ZZ
Y ---> 0
Z ---> 1

Answer (Please select your correct option)

- 110
- 11011
- 1101111
- 11101

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Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following refers to an effectively solvable problem having answer in "Yes" or "No"?

Answer (Please select your correct option)

- Decision procedure
- Logical problem
- Analytical problem
- Binary procedure

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Question No : 33 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular

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bhuraa

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Question No : 32 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ and _____ are removed in order to make a CFG in Chomsky Normal Form (CNF).

Answer (Please select your correct option)

- Null, nullable productions
- Nullable, unit productions
- Null, unit productions
- Strings of length 0, null

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Question No : 31 of 52 Marks: 1 (Budgeted Time 1 Min)

A language of strings of prime length is considered to be:

Answer (Please select your correct option)

- regular but infinite
- regular
- irregular
- regular but finite

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Question No : 30 of 52 Marks: 1 (Budgeted Time 1 Min)

If we have two transiton graphs then their union will be expressed by:

Answer (Please select your correct option)

- taking a common start state and joining them by two null transitions
- just connecting both start states by null transitions
- connecting final state of first TG to the initial state of second TG
- connecting the final state of first TG to the final state of second TG

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Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Language of strings having even number of a's is considered to be:

Answer (Please select your correct option)

- Regular
- Irrelgular
- Irregular and infinite
- Irregular and finite

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG) and select the correct choice
S→AB
A →BSB | a | b
B→CC
C→SS | b | bb

Answer (Please select your correct option)

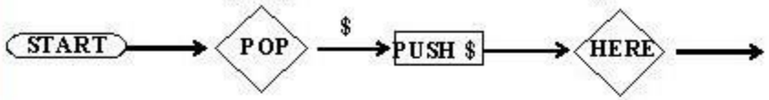
- abb is a word in the corresponding CFL.
- abb is not the word of corresponding CFL.
- Any word can be accepted from the corresponding CFL.
- b is a word in the corresponding CFL.

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

Look at the following diagram and select the correct option:



Answer (Please select your correct option)

- The PDA must not begin with the sequence
- The PDA may begin with the sequence
- The PDA must begin with the sequence
- The PDA may not begin with the sequence

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Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For every Context Free Grammar (CFG), we can make the corresponding _____.

Answer (Please select your correct option)

- FA
- TG
- PDA
- Regular Grammar

101% sure :D :D but not sure

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer (Please select your correct option)

- r1* r2 r3*
- (r1r2r3)*
- r1*r2*r3*
- None of the given

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there must be transition for all letters.

Answer (Please select your correct option)

- NFA
- FA
- TG
- GTG

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL)
- Regular
- Finite

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Question No : 19 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Σ={a,Aa,Abb}, then string aAaAbbAa has length equal to_____.

Answer (Please select your correct option)

- Three
- Four
- One
- Two

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

There are different parsing techniques, including _____

Answer (Please select your correct option)

- Top down parsing
- Bottom up parsing
- Parsing technique for particular grammar of arithmetic expression
- All of these

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The process of finding the derivation of word generated by particular grammar is called_____

Answer (Please select your correct option)

- Processing
- Parsing
- Programming
- Planing

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Question No : 16 of 52 Marks: 1 (Budgeted Time 1 Min)

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

TM is more powerful than FSM because

Answer (Please select your correct option)

- The tape movement is confined to one direction
- It has no finite state control
- It has the capability to remember arbitrary long sequences of input symbols
- The tape movement is confined to more than one directions

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Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following statements is INCORRECT:

Answer (Please select your correct option)

- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these
- (a+b)*aa(a+b)* generates Regular language.

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Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Regular languages are closed under _____

Answer (Please select your correct option)

- Union and Concatenation
- Union and Division
- Union and Exponent
- Union and Factorial

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Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

There exist _____ derivations of a certain word in ambiguous grammar.

Answer (Please select your correct option)

- more than one
- Exactly one
- exactly two
- no

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Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following pairs of regular expressions are not equivalent?

Answer (Please select your correct option)

- (ab)* and a*b*
- x(xx)* and (xx)*x
- x^+ and x^*x^+
- x(x)* and (x)*x

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Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

An alphabet of Σ is valid if

Answer (Please select your correct option)

- No letter of Σ appears in middle of any other letter
- No letter of Σ appears at end of any other letter
- No letter of Σ appears at start of any other letter
- No letter of Σ appears at end or middle of any other letter

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following pairs of regular expressions are equivalent?

Answer (Please select your correct option)

- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x^+ and x*
- x^+ and x*x^+

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

Answer (Please select your correct option)

- All
- No
- One
- Two

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Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If L1 and L2 are two regular languages, then _____ is/are also regular.

Answer (Please select your correct option)

- Both $L1 \cup L2$, $L1 \cap L2$
- $L1^c \cap L2$
- $L1 \cup L2$
- $L1 \cap L2$

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Question No : 6 of 52 Marks: 1 (Budgeted Time 1 Min)

Even-palindrome is a _____ language

Answer (Please select your correct option)

- non-regular
- regular
- regular but infinite
- regular but finite

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Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

"One GTG can generate _____ language".

Answer (Please select your correct option)

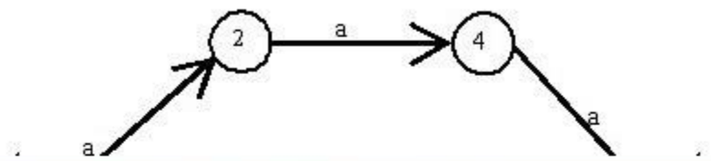
- more than one
- only one
- only two
- only three

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Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given TG's are _____.



Answer (Please select your correct option)

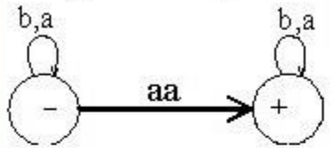
- non-symmetric
- Equivalent
- Non-equivalent
- TG's are not valid

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Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given TG has _____ RE.



Answer (Please select your correct option)

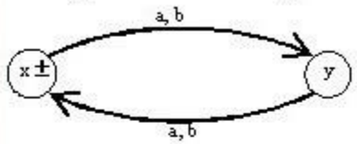
- (b+a) (aa) (b+a)*
- (a+b) (aa) (a+b)*
- (a+b)* (aa) (a+b)*
- (a+b)* (aa) (a+b)

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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA can be represented by



Answer (Please select your correct option)

- ((a+ b) (a + b))*
- (a + b)(a + b)*
- (a + b)(a + b)
- (a + b)*(a + b)*

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Question No : 9 of 52 Marks: 1 (Budgeted Time 1 Min)

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Answer (Please select your correct option)

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- x^+ and x^*x^+

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

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- One
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- $L1^c \cap L2$
- $L1 \cup L2$
- $L1 \cap L2$

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Even-palindrome is a _____ language

Answer (Please select your correct option)

- non-regular
- regular
- regular but infinite
- regular but finite

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Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

"One GTG can generate _____ language".

Answer (Please select your correct option)

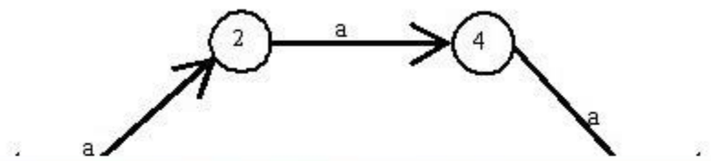
- more than one
- only one
- only two
- only three

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Question No : 4 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given TG's are _____.



Answer (Please select your correct option)

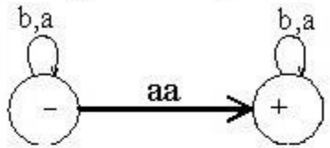
- non-symmetric
- Equivalent
- Non-equivalent
- TG's are not valid

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Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given TG has _____ RE.



Answer (Please select your correct option)

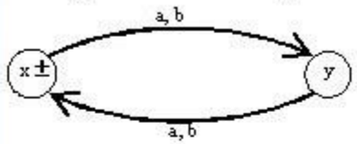
- (b+a) (aa) (b+a)*
- (a+b) (aa) (a+b)*
- (a+b)* (aa) (a+b)*
- (a+b)* (aa) (a+b)

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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA can be represented by



Answer (Please select your correct option)

- ((a+ b) (a + b))*
- (a + b)(a + b)*
- (a + b)(a + b)
- (a + b)*(a + b)*

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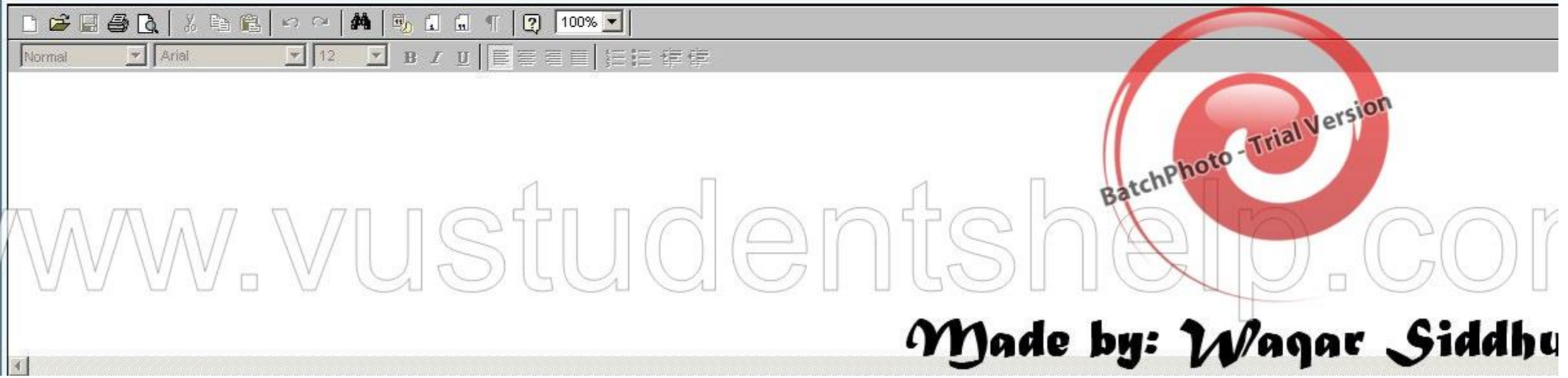
Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

a. Construct Context Free Grammar (CFG) for the language defined by the regular expression:

(ab + ba)*(a + b)*

b. Remove null productions from the CFG constructed in part i.

Answer (Please click here to Add Answer)

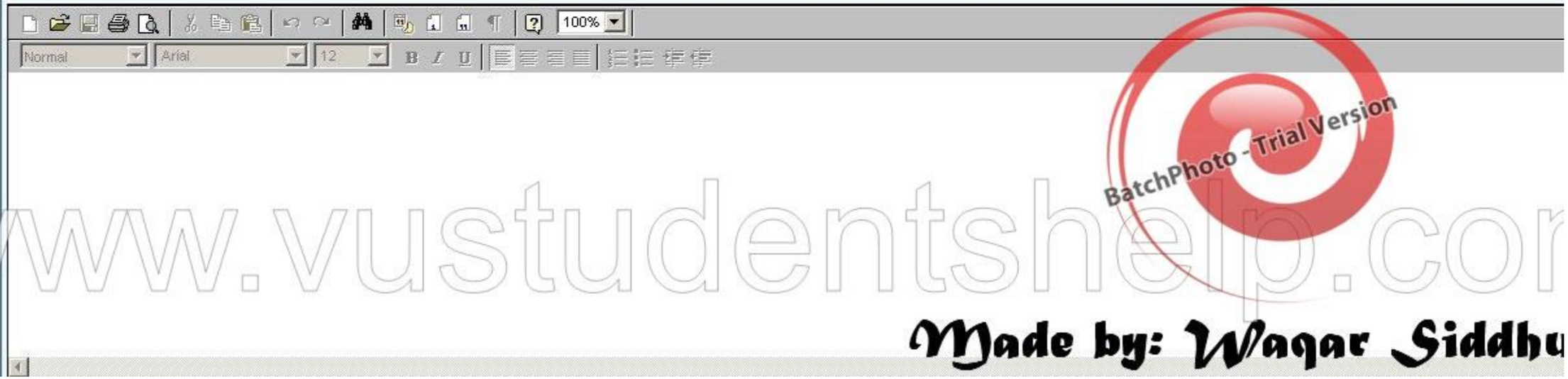


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Question No : 51 of 52 Marks: 5 (Budgeted Time 10 Min)

Using suitable examples, differentiate between Kleene star closure and plus operation?

Answer (Please click here to Add Answer)

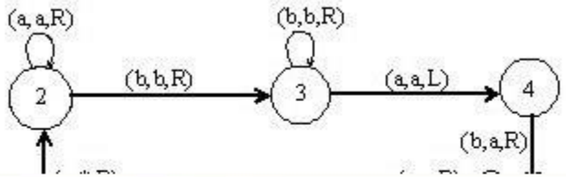


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Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the following TM



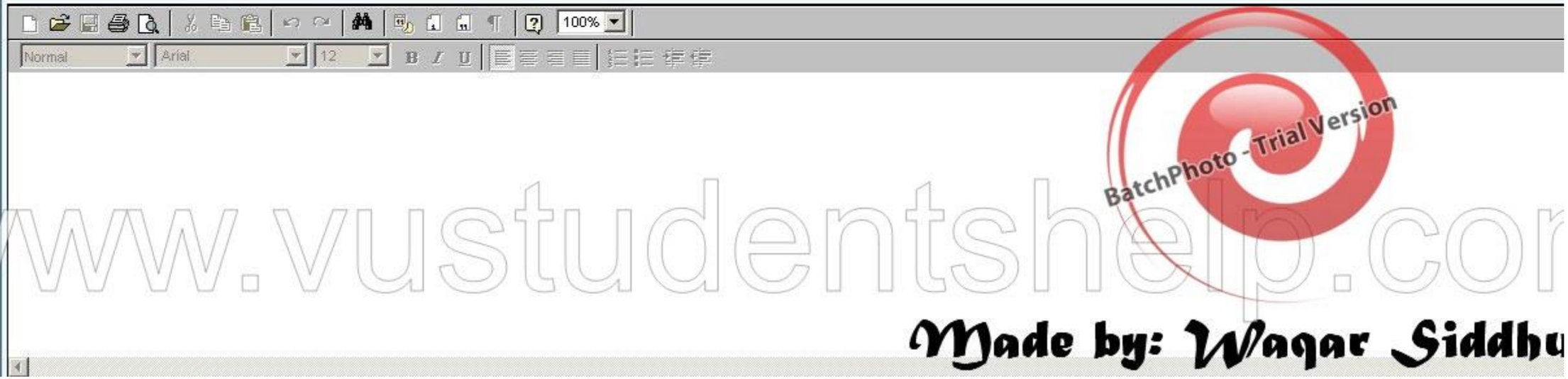
Answer (Please click here to Add Answer)

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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Write the CFG for EVEN-EVEN language?

Answer (Please click here to Add Answer)

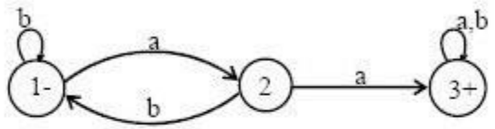


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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Let us consider the following Finite Automaton (FA):



Answer (Please click here to Add Answer)

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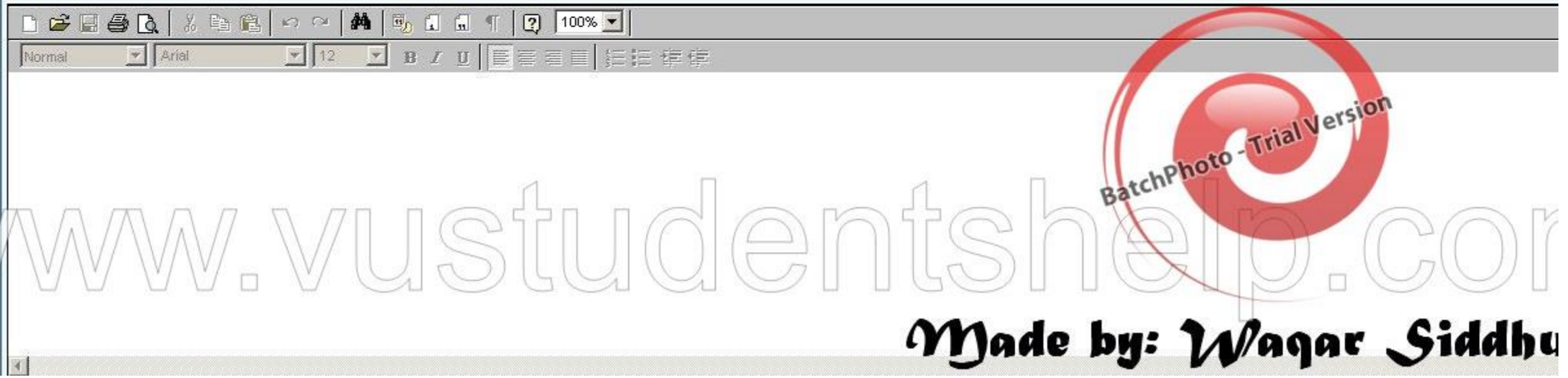
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Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Write down Context Free Grammar (CFG) for the language of strings expressed by the following regular expression:

$(a + b)^*(a + b)^+$

Answer (Please click here to Add Answer)



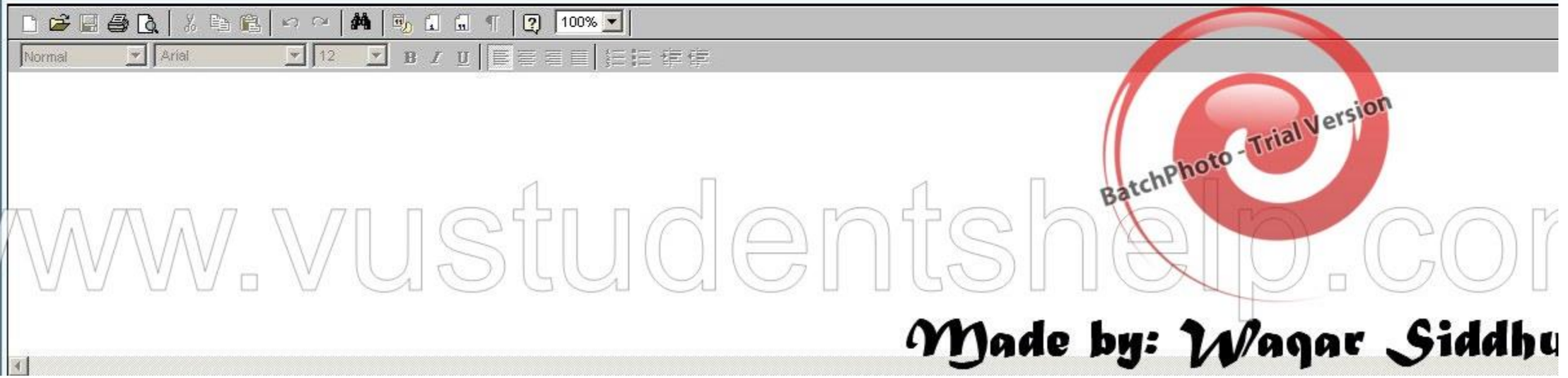
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Question No : 46 of 52 **Marks: 3 (Budgeted Time 6 Min)**

What does the following arbitrary summary table show?

FROM Where	TO Where	READ What	POP What	PUSH What	ROW Number
$READ_0$	$READ_3$	b	b	abb	11

Answer (Please click here to Add Answer)



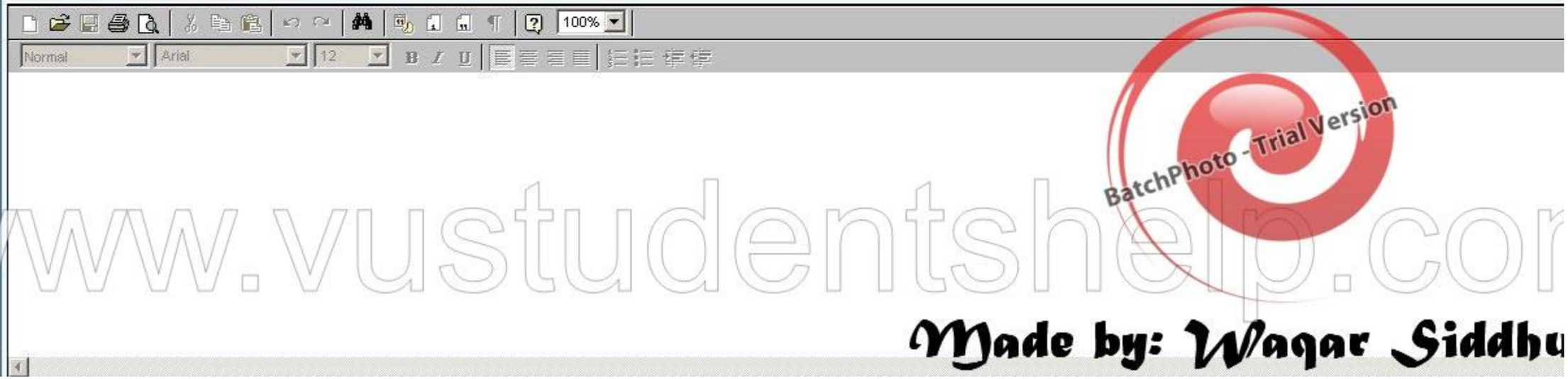
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Question No : 45 of 52 Marks: 3 (Budgeted Time 6 Min)

What are the steps of Recursive Definition of Languages?

Answer (Please click here to Add Answer)

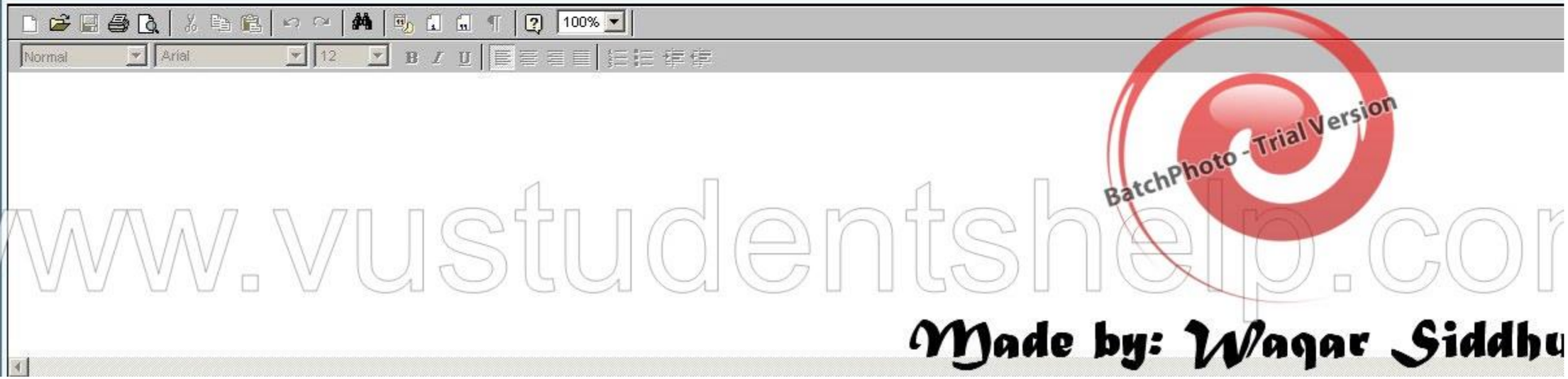


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Question No : 44 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Write down regular expression for a language L of strings defined over Σ = {a, b} that neither ends in "a" nor with "b".

Answer (Please click here to Add Answer)

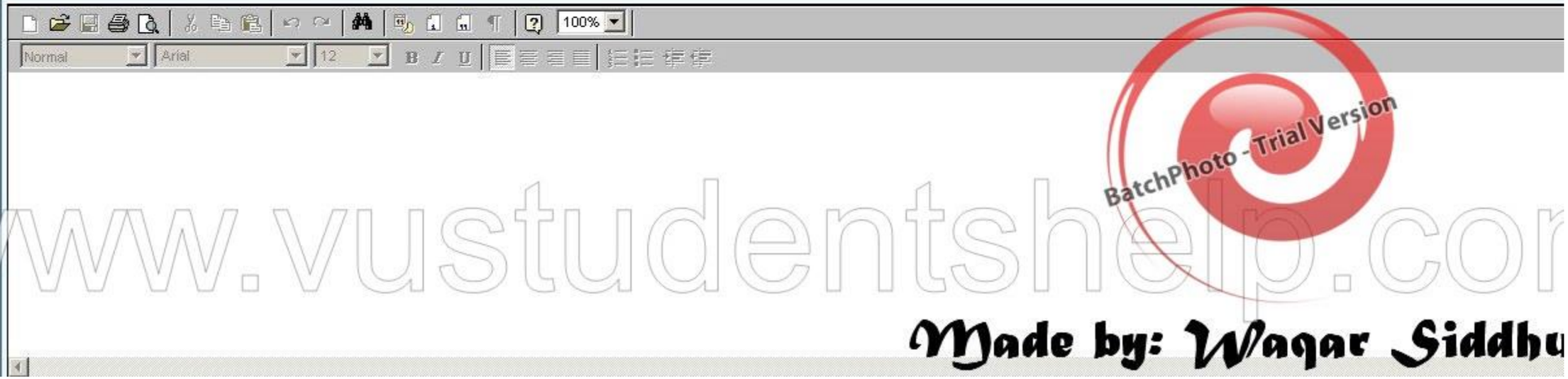


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Question No : 43 of 52 Marks: 2 (Budgeted Time 4 Min)

What is meant by Row language?

Answer (Please click here to Add Answer)



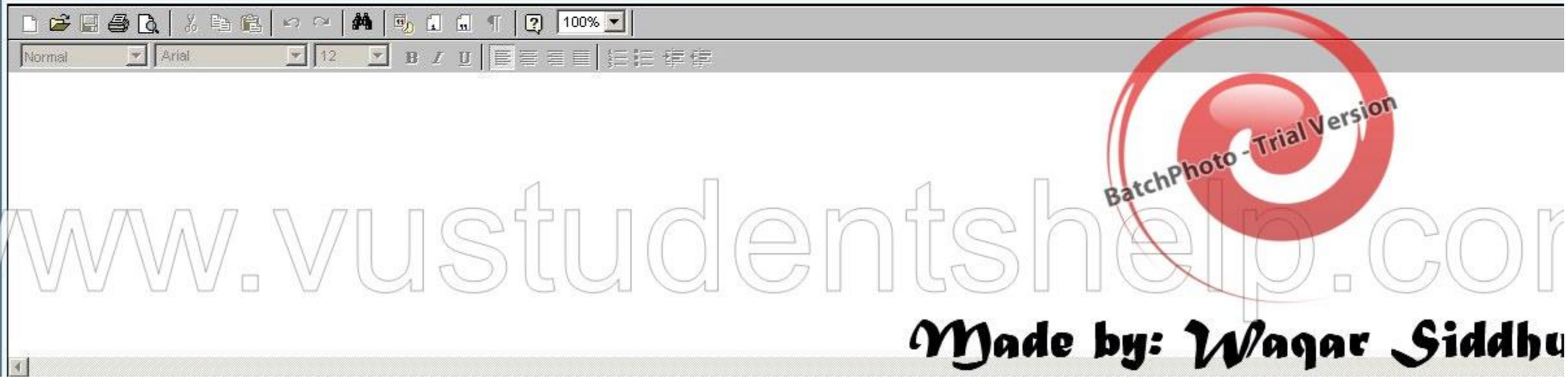
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Question No : 42 of 52 Marks: 2 (Budgeted Time 4 Min)

What are the uses of Push Down Automata (PDA) in computing?

Answer (Please click here to Add Answer)



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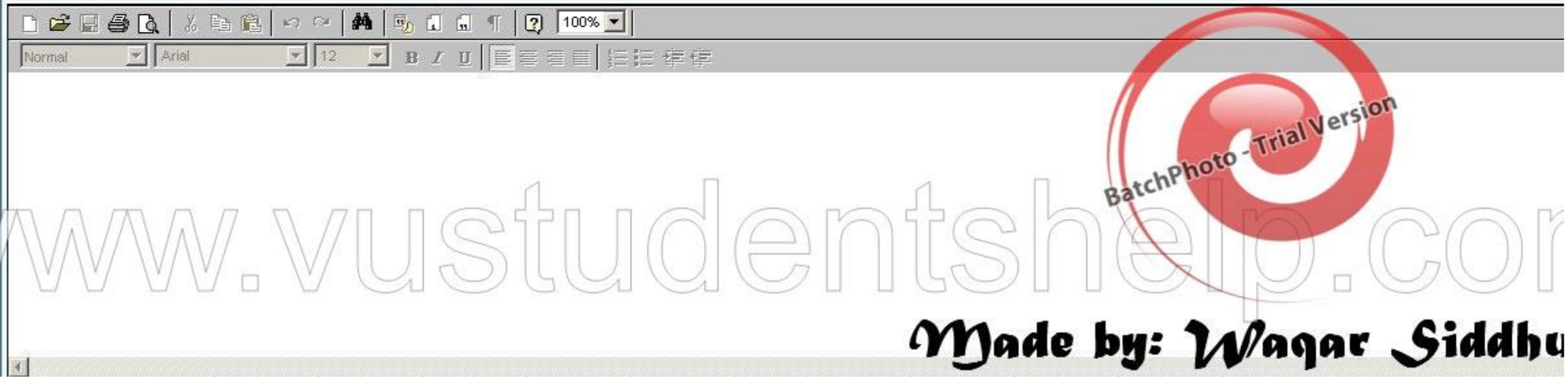
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Question No : 41 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Write alternative form of this production.

$S \rightarrow aS,\ S \rightarrow bS,\ S \rightarrow \Lambda$

Answer (Please click here to Add Answer)



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Question No : 40 of 52 Marks: 1 (Budgeted Time 1 Min)

_____ is always Deterministic.

Answer (Please select your correct option)

- Finite Automaton
- Transition Graph
- Generalize Transition Graph
- Non-deterministic finite automaton

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Question No : 39 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A nonterminal is said to be self-embedded, if in a given derivation of a word, it ever occurs as a tree _____ of itself.

Answer (Please select your correct option)

- ascendant
- descendant
- top
- bottom

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Question No : 38 of 52 Marks: 1 (Budgeted Time 1 Min)

Concatenation of r1 and r2 will be shown as:

Answer (Please select your correct option)

- r1+r2
- r2+r1
- r2r1
- r1r2

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Question No : 37 of 52 Marks: 1 (Budgeted Time 1 Min)

Which of the following is a regular Context Free Grammar:

Answer (Please select your correct option)

- S→abS | baS | ^
- S→aSb | baS |^
- S→abS | bSa | ^
- S→aSb | Sa | ^

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S → SX|XY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

- 001
- 00010
- 0010000
- 00100

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S ---> SX|XY
X ---> ZZ
Y ---> 0
Z ---> 1

Answer (Please select your correct option)

- 110
- 11011
- 1101111
- 11101

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Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following refers to an effectively solvable problem having answer in "Yes" or "No"?

Answer (Please select your correct option)

- Decision procedure
- Logical problem
- Analytical problem
- Binary procedure

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Question No : 33 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular

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Question No : 32 of 52 **Marks: 1 (Budgeted Time 1 Min)**

_____ and _____ are removed in order to make a CFG in Chomsky Normal Form (CNF).

Answer (Please select your correct option)

- Null, nullable productions
- Nullable, unit productions
- Null, unit productions
- Strings of length 0, null

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Question No : 31 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A language of strings of prime length is considered to be:

Answer (Please select your correct option)

- regular but infinite
- regular
- irregular
- regular but finite

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Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If we have two transiton graphs then their union will be expressed by:

Answer (Please select your correct option)

- taking a common start state and joining them by two null transitions
- just connecting both start states by null transitions
- connecting final state of first TG to the initial state of second TG
- connecting the final state of first TG to the final state of second TG

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Question No : 29 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Language of strings having even number of a's is considered to be:

Answer (Please select your correct option)

- Regular
- Irrelgular
- Irregular and infinite
- Irregular and finite

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG) and select the correct choice
S→AB
A →BSB | a | b
B→CC
C→SS | b | bb

Answer (Please select your correct option)

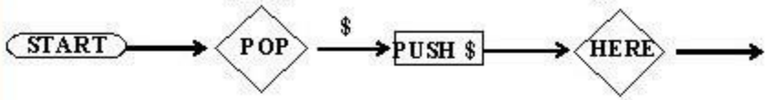
- abb is a word in the corresponding CFL.
- abb is not the word of corresponding CFL.
- Any word can be accepted from the corresponding CFL.
- b is a word in the corresponding CFL.

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Question No : 26 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Look at the following diagram and select the correct option:



Answer (Please select your correct option)

- The PDA must not begin with the sequence
- The PDA may begin with the sequence
- The PDA must begin with the sequence
- The PDA may not begin with the sequence

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Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

For every Context Free Grammar (CFG), we can make the corresponding _____.

Answer (Please select your correct option)

- FA
- TG
- PDA
- Regular Grammar

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

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Question No : 23 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 22 of 52 Marks: 1 (Budgeted Time 1 Min)



Answer (Please select your correct option)

- r1* r2 r3*
- (r1r2r3)*
- r1*r2*r3*
- None of the given

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

In _____ there must be transition for all letters.

Answer (Please select your correct option)

- NFA
- FA
- TG
- GTG

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL)
- Regular
- Finite

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Question No : 19 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ={a,Aa,Abb}, then string aAaAbbAa has length equal to_____.

Answer (Please select your correct option)

- Three
- Four
- One
- Two

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Question No : 18 of 52 Marks: 1 (Budgeted Time 1 Min)

There are different parsing techniques, including _____

Answer (Please select your correct option)

- Top down parsing
- Bottom up parsing
- Parsing technique for particular grammar of arithmetic expression
- All of these

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Question No : 17 of 52 Marks: 1 (Budgeted Time 1 Min)

The process of finding the derivation of word generated by particular grammar is called_____

Answer (Please select your correct option)

- Processing
- Parsing
- Programming
- Planing

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Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 15 of 52 Marks: 1 (Budgeted Time 1 Min)

TM is more powerful than FSM because

Answer (Please select your correct option)

- The tape movement is confined to one direction
- It has no finite state control
- It has the capability to remember arbitrary long sequences of input symbols
- The tape movement is confined to more than one directions

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Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following statements is INCORRECT:

Answer (Please select your correct option)

- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these
- (a+b)*aa(a+b)* generates Regular language.

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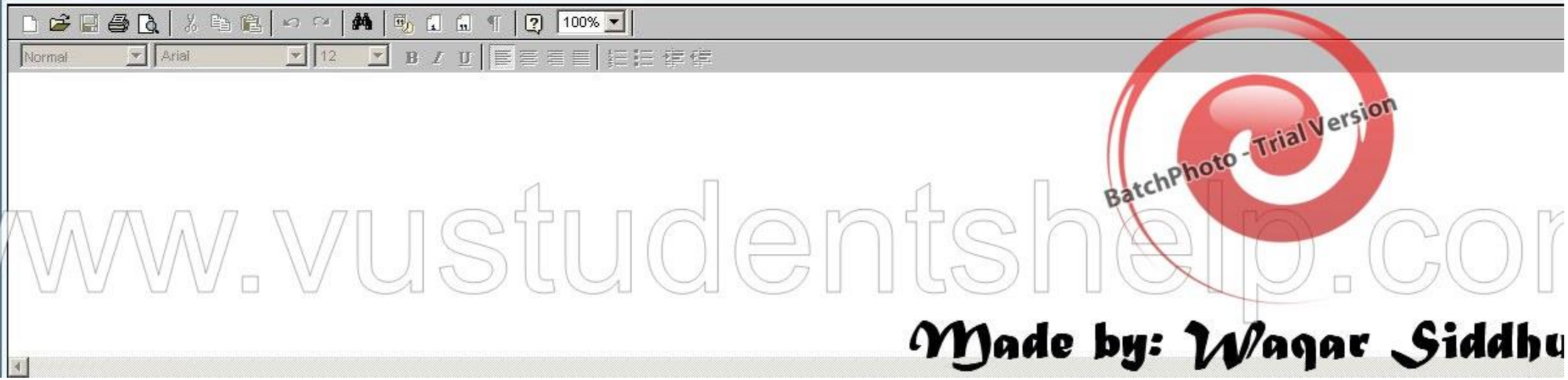
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Question No : 52 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Suppose a Maze of 20 boxes; 1 box for initial and 1 for final state. 6 boxes are for non-moveable states while the rest can be used for transitions. Draw a Nondeterministic Finite Automaton (NFA) fo the given Maze.

Marks [2.5+2.5]

Answer (Please click here to Add Answer)



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Question No : 51 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Derive Chomsky Normal Form (CNF) corresponding to the Context Free Grammar (CFG) given below:

$A \rightarrow BAB \mid B \mid \Lambda$

$B \rightarrow 00 \mid \Lambda$

Answer (Please click here to Add Answer)

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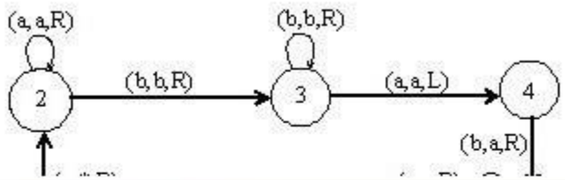
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Question No : 50 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the following TM



Answer (Please click here to Add Answer)

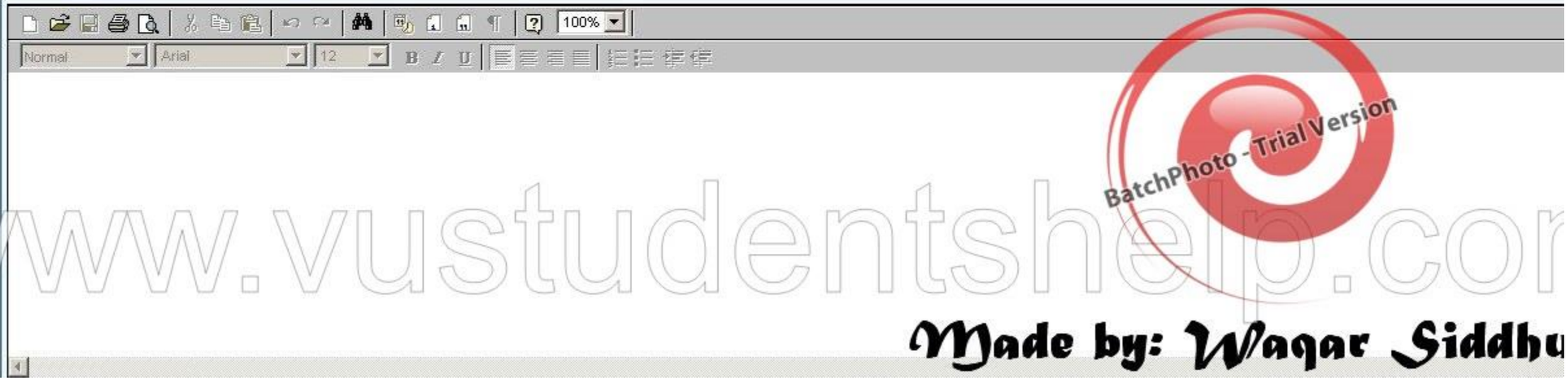
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Question No : 49 of 52 **Marks: 5 (Budgeted Time 10 Min)**

Consider the language L which is EVEN-EVEN, defined over $\Sigma = \{a,b\}$. In how many classes does L may partition Σ^*. Explain briefly.

Answer (Please click here to Add Answer)

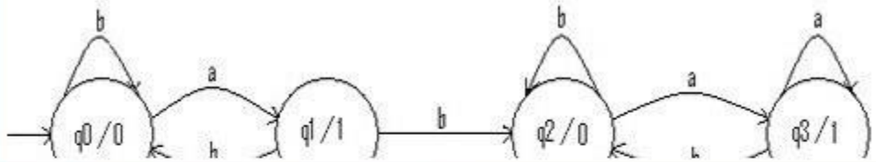


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Question No : 48 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Consider the following Moore machine:



Answer (Please click here to Add Answer)

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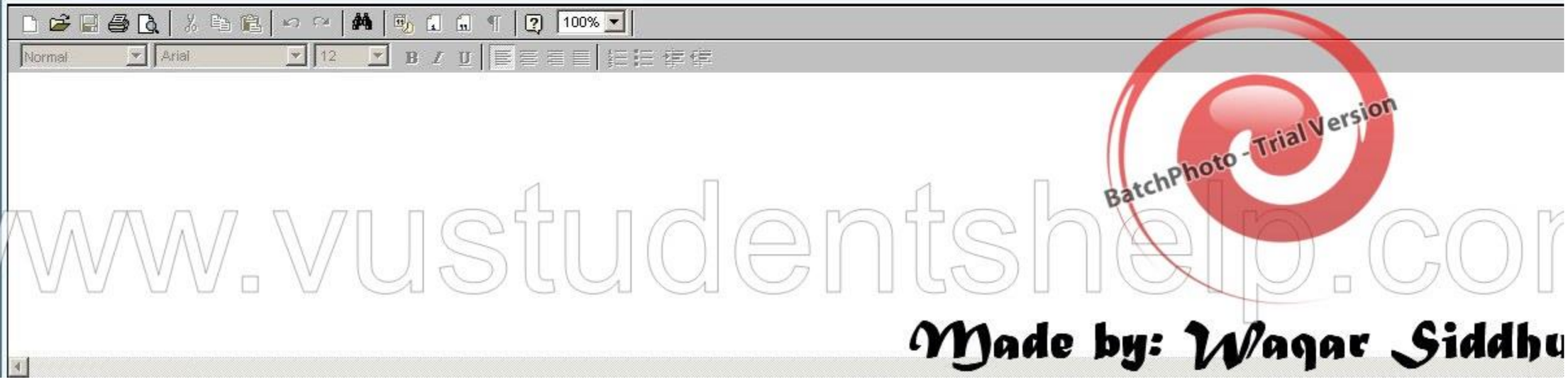
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Question No : 47 of 52 **Marks: 3 (Budgeted Time 6 Min)**

Illustrate Decidable Problem with the help of an example.

Marks [1+2]

Answer (Please click here to Add Answer)



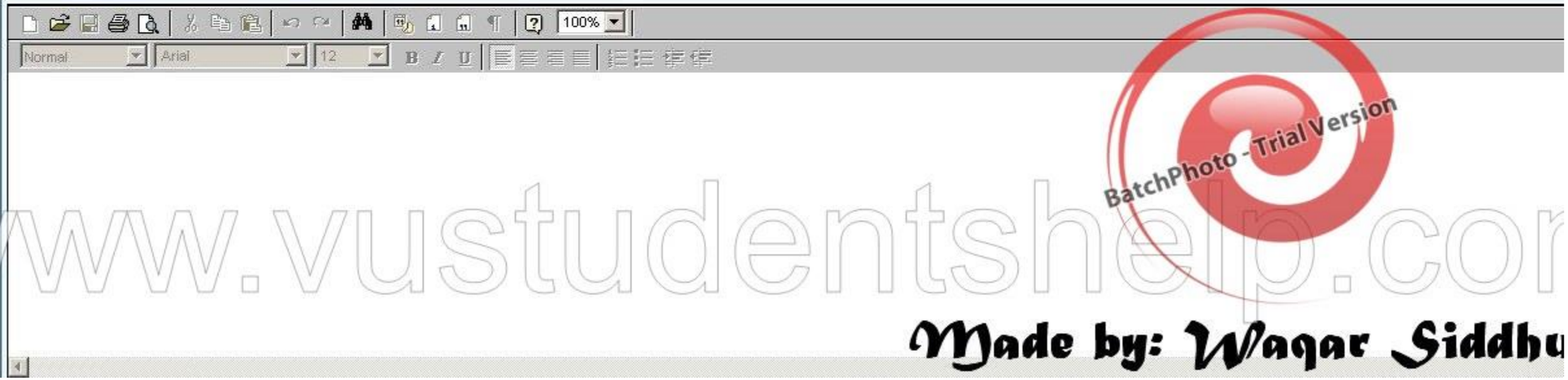
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Question No : 46 of 52 Marks: 3 (Budgeted Time 6 Min)

Write three different parsing techniques.

Answer (Please click here to Add Answer)



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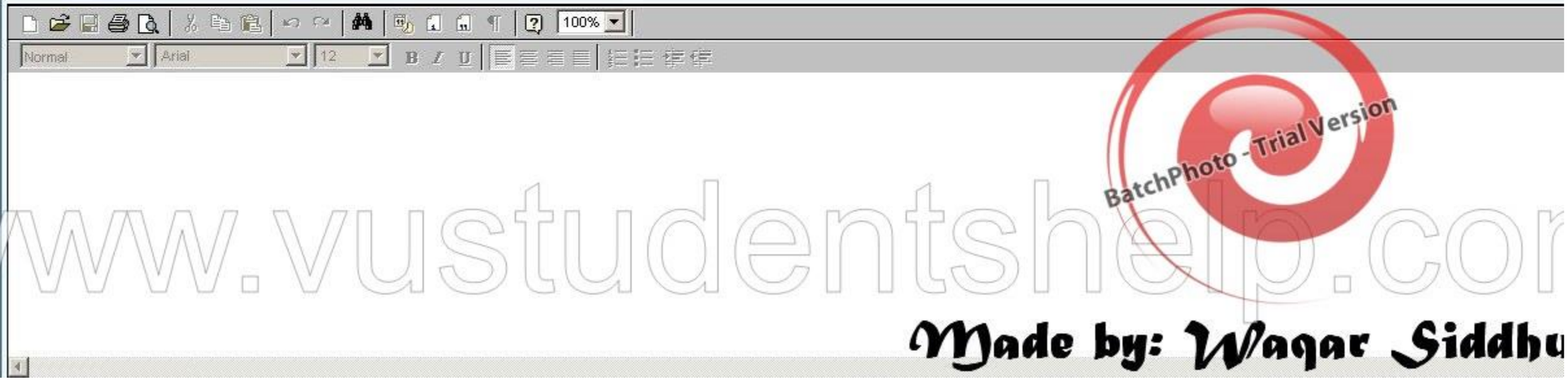
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Question No : 45 of 52 **Marks: 3 (Budgeted Time 6 Min)**

What is meant by the following statement ?

"LANGUAGE IS CLOSED"

Answer (Please click here to Add Answer)



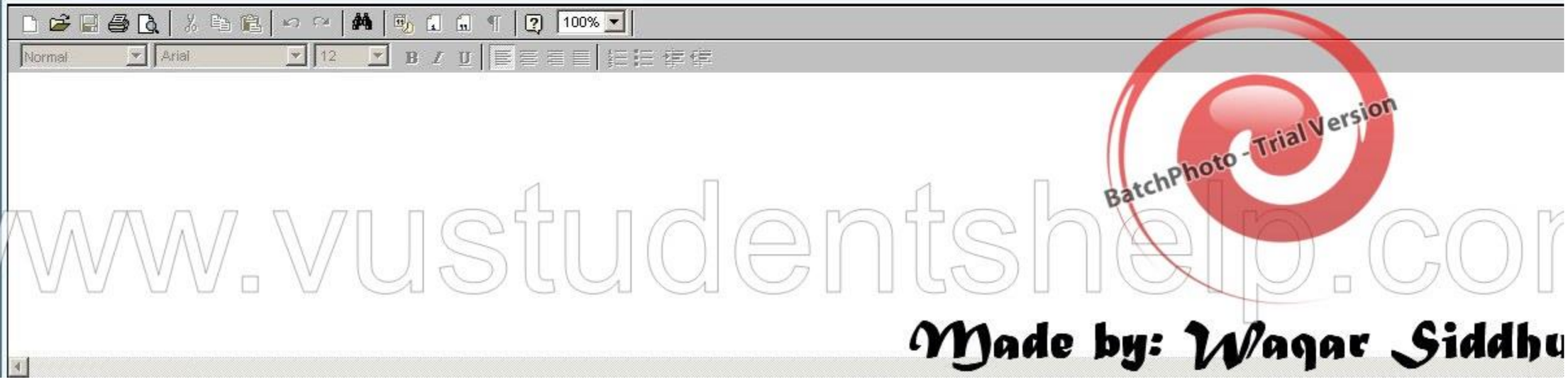
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Question No : 44 of 52 Marks: 2 (Budgeted Time 4 Min)

Consider live productions to be applied during derivation of a word. In your point of view what will be the ultimate result?

Answer (Please click here to Add Answer)



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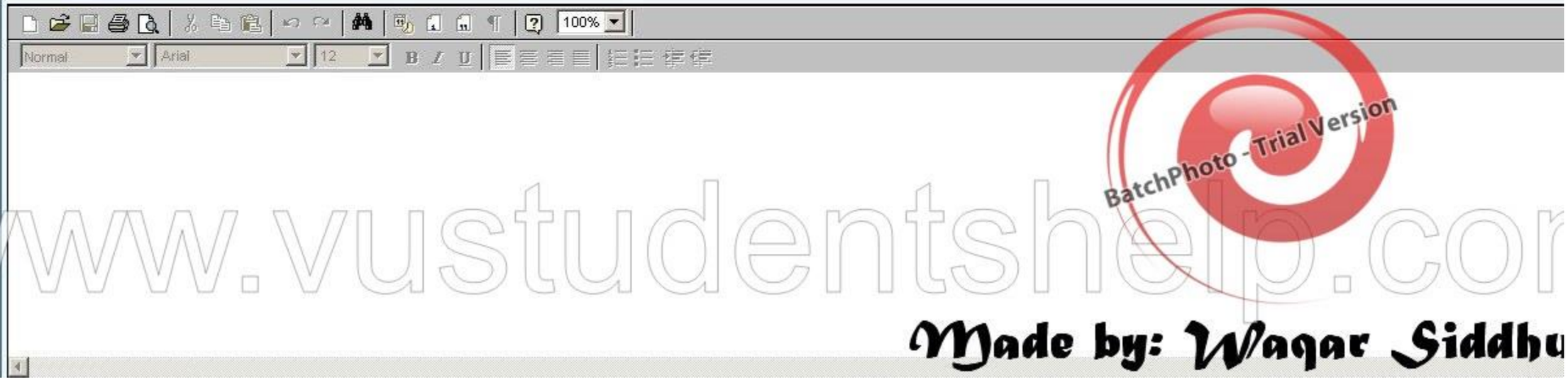
Question No : 43 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Consider the language defined by the following regular expression:

b(a + b)*a

What will be the corresponding Context Free Grammar (CFG)?

Answer (Please click here to Add Answer)



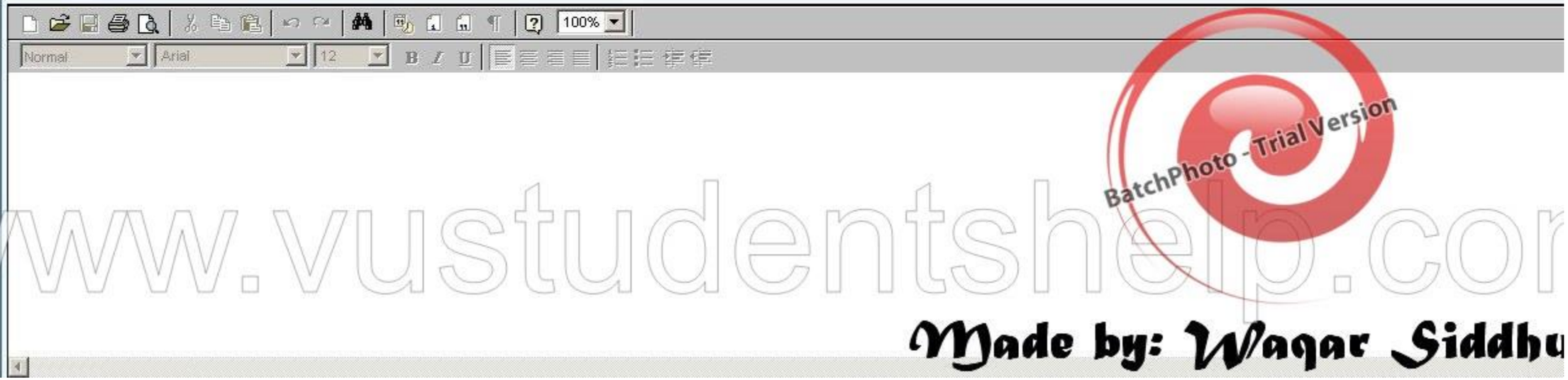
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Question No : 42 of 52 **Marks: 2 (Budgeted Time 4 Min)**

Explain the language L of string, defined over ∑= {0,1}, having double 0's or double 1's?

Answer (Please click here to Add Answer)



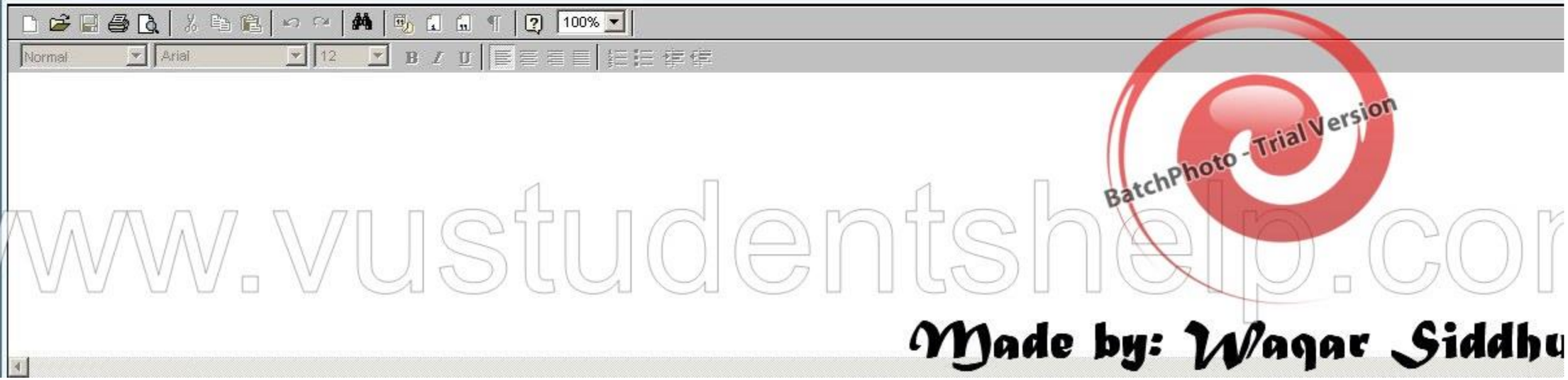
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Question No : 41 of 52 **Marks: 2 (Budgeted Time 4 Min)**

What is meant by "Left most Derivation" in context of CFG?

Answer (Please click here to Add Answer)



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Question No : 40 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Let us consider the following Turing machine (™):



Answer (Please select your correct option)

- fffgggg
- fffggggf
- fffggg
- ffffggg

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Question No : 39 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S---->01A

A---->10B

B---->0A|11

Answer (Please select your correct option)

- 011011
- 011011011
- 011001011
- 011001001011

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Question No : 38 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following states do not have outgoing edges:

Answer (Please select your correct option)

- Start
- Read
- Pop
- Accept

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Question No : 37 of 52 **Marks: 1 (Budgeted Time 1 Min)**

A read state can have _____ outgoing edge/ edges.

Answer (Please select your correct option)

- 1
- 2
- 3
- Any number of

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Question No : 36 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following Context Free Grammar (CFG):

S → SX|XY
X → ZZ
Y → 1
Z → 0

Answer (Please select your correct option)

- 00100
- 0010000
- 001
- 00010

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Question No : 35 of 52 **Marks: 1 (Budgeted Time 1 Min)**

The languages are supposed to be _____ in case of testing pumping lemma.

Answer (Please select your correct option)

- context free
- irregular
- both regular and irregular
- regular

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Question No : 34 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following theorem partitions Σ^* into distinct classes?

Answer (Please select your correct option)

- Decidability theorem
- Myhill Nerode theorem
- Kleen's theorem
- Emptiness theorem

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Question No : 33 of 52 Marks: 1 (Budgeted Time 1 Min)

The language of $a^n b^n$ can be expressed by:

Answer (Please select your correct option)

- FA
- TG
- GTG
- PDA

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Question No : 32 of 52 **Marks: 1 (Budgeted Time 1 Min)**

We can have more than one parse trees for a given string in case of _____ Context Free Grammar (CFG).

Answer (Please select your correct option)

- vague
- ambiguous
- hazy
- simple

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Question No : 31 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement(s) is/are true or false?

1) The turing machine is similar to a finite automaton but with an unlimited and unrestricted memory.
2) A Turing machine is a much more accurate model of a general purpose computer.

Answer (Please select your correct option)

- Statement 1 is true
- Statement 2 is true
- Both statements (1 & 2) are false
- Statement 2 is false

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Question No : 30 of 52 **Marks: 1 (Budgeted Time 1 Min)**

In _____ there are Finite set of states containing exactly one START state and some (may be none) HALT states that cause execution to terminate, when the HALT states are entered.

Answer (Please select your correct option)

- TG
- Turing Machine
- GTG
- Finite Automata

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Question No : 29 of 52 Marks: 1 (Budgeted Time 1 Min)

Can a turing machine's head ever be in the same location in two successive steps?

Answer (Please select your correct option)

- Yes
- Yes but only in finite languages
- No
- Yes but only in infinite langauges

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Question No : 28 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If a given Context Free grammar (CFG) contains a production of the form S→Λ, then _____

Answer (Please select your correct option)

- The corresponding CFL is not empty.
- The corresponding CFL is empty.
- The corresponding CFL is not empty if the language is regular
- The corresponding CFL is empty and non regular.

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Question No : 27 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If a language can be expressed by a regular expression, then its complement is _____ .

Answer (Please select your correct option)

- Regular
- Non Regular
- Finite
- Infinite

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Question No : 26 of 52 Marks: 1 (Budgeted Time 1 Min)

A regular language _____ .

Answer (Please select your correct option)

- Must be finite
- Must be infinite
- Can be finite or infinite
- Complex Language

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Question No : 25 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Left hand side of a production in Conext free grammer (CFG) consists of _____.

Answer (Please select your correct option)

- One terminal
- More than one terminal
- One non-terminal
- Terminals and non-terminals

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Question No : 24 of 52 Marks: 1 (Budgeted Time 1 Min)

The _____ machine helps in building a machine that can perform the addition of binary numbers.

Answer (Please select your correct option)

- Incrementing
- Complementing
- Decrementing
- Multiplication

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Question No : 23 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following is RE for a language defined over Σ={a,b} having words starting with a:

Answer (Please select your correct option)

- (a + b)*
- aa(a + b)+
- a(a + b)*
- a*(a + b)

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Question No : 22 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If r1 and r2 are regular expressions then (r1) (r2) is a _____ .

Answer (Please select your correct option)

- RE
- TG
- FA
- GTG

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Question No : 21 of 52 Marks: 1 (Budgeted Time 1 Min)

The language generated by CFG is called _____.

Answer (Please select your correct option)

- Infinite
- Context Free Language (CFL)
- Regular
- Finite

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Question No : 20 of 52 Marks: 1 (Budgeted Time 1 Min)

Σ= {a,b} then string abbabb has length equal to _____.

Answer (Please select your correct option)

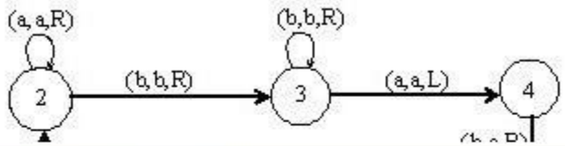
- Three
- Four
- Five
- Six

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Question No : 19 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the following TM



Answer (Please select your correct option)

- Above TM accepts the non-CFL {a b c}.
- Above TM accepts the non-CFL $\{a^n b^n a^n\}$.
- Above TM accepts the non-CFL $\{a^n b^{n+2} a^n\}$.
- Above TM accepts the non-CFL $\{a^{n+2} b^n a^n\}$.

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Question No : 18 of 52 **Marks: 1 (Budgeted Time 1 Min)**

CYK algorithm was invented by John Cocke and later was published by ____

Answer (Please select your correct option)

- Talik Khon and Daniel H
- Tandao Kasami and Daniel H
- Tandao Dagger and Daniel H
- Talik Khon

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Question No : 17 of 52 **Marks: 1 (Budgeted Time 1 Min)**

We can also represent an FA using different states e.g. Accept state, Reject state, Read State etc.
The _____ state behaves as final state of an FA.

Answer (Please select your correct option)

- Accept
- Pop
- Push
- Reject

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Question No : 16 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which one of the following statements is INCORRECT:

Answer (Please select your correct option)

- (a+b)*aa(a+b)* generates Regular language.
- A language consisting of all strings over ∑={a,b} having equal number of a's and b's is a regular language
- Every language that can be expressed by FA can also be expressed by RE
- None of these

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Question No : 15 of 52 **Marks: 1 (Budgeted Time 1 Min)**

There exist _____ derivations of a certain word in ambiguous grammar.

Answer (Please select your correct option)

- more than one
- Exactly one
- exactly two
- no

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Question No : 14 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following statement is correct?

Answer (Please select your correct option)

- The length of output string is greater than length of input string in moore machine.
- The length of output string is greater than length of input string in mealy machine.
- The length of output string is equal to length of input string in moore machine.
- The length of output string is less than length of input string in mealy machine.

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Question No : 13 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following pairs of regular expressions are not equivalent?

Answer (Please select your correct option)

- (ab)* and a*b*
- x(xx)* and (xx)*x
- x^+ and x^*x^+
- x(x)* and (x)*x

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Question No : 12 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Which of the following pairs of regular expressions are equivalent?

Answer (Please select your correct option)

- 1(001)* and (10)*10
- x(xx)* and (x)*x
- x^+ and x^*
- x^+ and x^*x^+

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Question No : 11 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If the FA has N states, then test the words of length less than N. If any one word is accepted by this FA, then it will accept ____ word.

Answer (Please select your correct option)

- All
- No
- One
- Two

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Question No : 10 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Pumping lemma is generally used to prove that:

Answer (Please select your correct option)

- A given language is infinite
- A given language is not regular
- Whether two given regular expressions of a regular language are equivalent or not
- A given language is finite

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Question No : 9 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Consider the language L of strings, defined over Σ = {a,b}, ending in a

Answer (Please select your correct option)

- There are finite many classes generated by L, so L is regular
- There are infinite many classes generated by L, so L is regular
- There are finite many classes generated by L, so L is non-regular
- There are infinite many classes generated by L, so L is non-regular

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Question No : 8 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If L1 and L2 are two regular languages, then _____ is/are also regular.

Answer (Please select your correct option)

- Both L1 ∪ L2 , L1 ∩ L2
- $L1^c \cap L2$
- L1 ∪ L2
- L1 ∩ L2

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Question No : 7 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Even-palindrome is a _____ language

Answer (Please select your correct option)

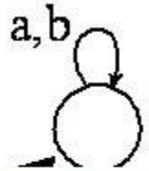
- non-regular
- regular
- regular but infinite
- regular but finite

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Question No : 6 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA accepts _____ language.



Answer (Please select your correct option)

- Finite
- Infinite
- No
- All

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Question No : 5 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Using tree structure, initial state can be represented by____

Answer (Please select your correct option)

- An arrow
- +
- -
- *

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Question No : 4 of 52 Marks: 1 (Budgeted Time 1 Min)

If an FA accepts no string, then language is denoted by _____

Answer (Please select your correct option)

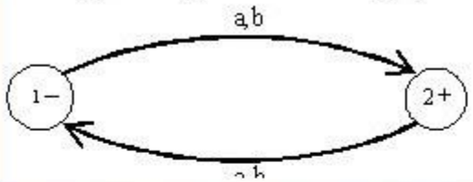
- {}
- Ø
- Both { } and Ø
- Null

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Question No : 3 of 52 **Marks: 1 (Budgeted Time 1 Min)**

Below given FA generates the language having strings of _____



Answer (Please select your correct option)

- ODD length
- EVEN length
- Equal number of a's and b's
- Odd number of a's and b's

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Question No : 2 of 52 **Marks: 1 (Budgeted Time 1 Min)**

If L1 and L2 are expressed by r1 and r2, respectively then the language expressed by (r1)(r2) will be _____

Answer (Please select your correct option)

- Regular
- Ir-regular
- Undermined
- Such language does not exist

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Question No : 1 of 52 Marks: 1 (Budgeted Time 1 Min)

Which statement is true?

Answer (Please select your correct option)

- All words are strings
- All strings are words
- Both are always same
- All letter are string

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